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An Analysis of Preimplantation Embryonic Death in Senescent Golden Hamsters[1-3]

THOMAS J. CONNORS[4], LARRY W. THORPE[5], AND A. L. SODERWALL

Department of Biology, University of Oregon, Eugene, Oregon 97403

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Young (3–6 month) and senescent (13–16 month) pregnant golden hamsters were killed 56 and 132 hr after ovulation. Ovaries and oviducts were serially sectioned in the 56-hr group. The uterine horns were bleached and cleared in the 132-hr group.

Embryonic mortality occurred in two stages. The number of corpora lutea was the same in both groups (7.3 $\pm$ 0.14 corpora lutea per ovary in young, 6.8 $\pm$ 0.19 corpora lutea per ovary in old). The number of ova surviving in the oviducts at 56 hr had decreased significantly ($p = <0.01$) to 4.9 $\pm$ 0.2 ova per oviduct in old animals compared with 6.6 $\pm$ 0.1 ova per oviduct in young.

The major source of embryonic mortality is failure of implantation in old animals. There were only 2.8 $\pm$ 1.5 implants per cornu in old hamsters and 6.1 $\pm$ 1.0 in young animals.

Female rodents enjoy a period of maximal reproductive output followed by a period of declining litter sizes (Biggers, Finn, and MacLaren, 1962). In hamsters, decreased fecundity begins at 14 months (Soderwall, Kent, Turbyfill, and Britenbaker, 1960) and is marked by a seven-fold increase in "preimplantation" death and a two-fold increase in resorption of established implantation sites (Thorneycroft and Soderwall, 1969).

Since the preimplantation mortality is defined by these authors as the reduction in number of grossly visible implants on the eighth day of pregnancy, the exact timing and cause of this reduction is not clear. It could involve several aspects including abnormal development, failute to implant, or, failure of early implantation sites to be maintained.

Blaha (1967) reported that senescent hamsters had fewer numbers of corpora lutea per cycle than young hamsters indicating a reduced ovulation rate in older animals. Thorneycroft and Soderwall (1969) found no difference in the ovulation rate in young or senescent hamsters during normal cycles and concluded that ovulation rate is of little significance in preimplantation death.

Blaha, in an earlier study (1964a), stated that implantation does not occur in many senescent hamsters and reported the presence of blastocysts in the uteri of senescent hamsters at a time when they should be implanted. No quantitative data were given, however, most evidence indicates a decreased responsiveness by the senile uterus. Blastocysts synchronously transferred from young donors to senescent recipients are less able

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[4] Present address: Department of Biological Sciences, Oregon Technical Institute, Klamath Falls, OR, 97601.

[5] Present address: Department of Obstetrics and Gynecology. University of Illinois at the Medical Center, Chicago, IL, 60680.

to implant successfully than those transferred to young recipients (Blaha, 1964b).

In addition to the decreased responsiveness of the senescent uterus, the ova from old hamsters are less viable than ova from young hamsters when transferred to the uteri of young recipients (Blaha, 1964b). The contribution of this reduced viability to preimplantation death has not been determined.

In order to investigate the nature and timing of preimplantation death reported by Thorneycroft and Soderwall (1969), the following study was done.

MATERIALS AND METHODS

The golden hamsters, *Mesocricetus auratus*, used in this study were taken from the colony maintained at the University of Oregon. The animals were kept in a room of constant 21C. temperature and were provided with 14 hr of light each day. Purina Laboratory Chow and lettuce greens given daily constituted the diet.

The young animals were 3–6 months of age and were virgin; the senescent animals were 13–16-months of age and were multiparous. Females in heat, as judged by the lordosis response (Soderwall, 1945) and vaginal discharge (Orsini, 1961) were placed with young males and copulations were observed. After one hour with the male, the females were examined for the presence of sperm in the vagina.

The timing used in this study is developmental age (fertilization at the time of ovulation) based upon 0100-hour ovulation the morning after initiation of behavioral estrus (Ward, 1948; Harvey, Yanagimachi, and Chang, 1964).

To determine the number of fertilized ova surviving in the oviduct young and senescent animals were killed 56 hr after ovulation. Their ovaries and oviducts were removed, fixed in Bouin's fluid for 24 hr, sectioned and stained with hematoxylin and eosin. Ovulation rate was determined by counting the number of corpora lutea in the ovary. The number of ova in the oviducts was counted in serial sections of the oviducts.

At 132 hr of development, young and senescent animals were killed, the uterine horns separated and fixed intact in alcohol-formalin-acetic acid solution. These uterine horns were bleached, dehydrated, and cleared by Orsini's method (1962a). The cleared uteri were viewed under oblique illumination and the implantation sites, which appear as discrete white opacities (Orsini, 1962b), were counted. The presence of these opacities was taken as *prima facie* evidence of successful implantation.

The statistical method used for differences between means was Student's *t* test using Bassel correction for small sample size (Dixon and Massey, 1957).

RESULTS

Both groups produced the same number of corpora lutea per ovary: 7.3 for young animals; 6.8 for old animals. The difference is not significant ($p > 0.05$).

By 56 hr developmental age, most ova are at the two-celled stage and only a few have completed the second cleavage division. There is no qualitative difference between the two groups. In both groups the ova are found clustered in the caudal end of the oviduct just anterior to the uterotubal junction.

There is a quantitative difference with senescent females averaging 4.9 ova per oviduct compared with 6.6 for young animals. The difference of 1.7 ova per oviduct is significant ($p < 0.01$).

The difference beween the number of corpora lutea and fertilized eggs in young animals is not significant. However, the difference of 1.9 ova per oviduct in the older group is significant ($p < 0.01$).

By 132 hr of normal development the blastocysts are implanted in the uterus and the implantation sites can be localized by viewing the cleared uteri under oblique illumination. The implantation sites are seen as discrete opaque areas and there is a qualitative difference between the two groups. The opaque areas in the older uteri are smaller and less developed than those of young animals. Figures 1 and 2 demonstrate these differences.

There is also a large difference in the number of implantation sites in the two groups. The young females implanted on the average, 6.1 blastocysts per uterine horn, whereas the older animals only implanted an average of 2.8. The difference of 3.3 implantation sites per cornu is significant ($p < 0.01$).

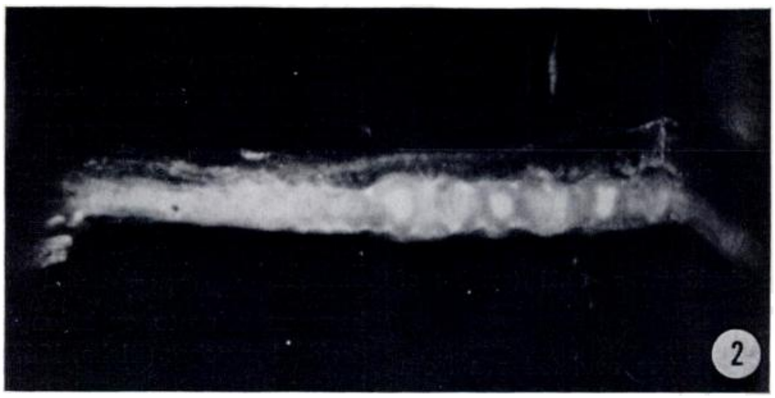

FIG. 1. Cleared uterine horn from a young animal at 132 hr after ovulation. The implantation sites are white opaque areas. ×2.5.

FIG. 2. Cleared uterine horn from an old animal at 132 hr after ovulation. The implantation sites are fewer, smaller and less developed than those in the young uterus. ×2.5.

The embryonic mortality data are summarized in Table 1.

DISCUSSION

The ovulation rate based upon the number of corpora lutea reported here is consistent with that previously reported for the hamster, 14.0 and 15.4 for senescent and young hamsters respectively (Thorneycroft and Soderwall, 1969). These findings appear in conflict with Blaha (1967) who found a reduction of corpora lutea in the ovaries of old hamsters.

Diminution of ovulation rate is not the cause of reproductive decline in other polytocuous rodents. There is no significant difference in ovulation rate between young (4–7 months) and old (10–13 months) mice (Biggers, Finn and MacLaren, 1962 and Harman and Talbert, 1967). In old rats, the number of corpora lutea remains normal although the mean litter size is smaller (Ingram, Mandel, and Zuckerman, 1958).

The difference in the number of ova surviving in the oviduct at 56 hr of development raises a question. Blaha (1964b) recovered fewer ova from flushings of old hamster uteri than from young uteri 63 and 68 hr after ovulation. He attributes this to a "decrease in the number of ova reaching the uterus in old animals," implying that some of the ova were left in the oviduct. The same phenomenon has been reported in old mice. Talbert and Krohn

TABLE 1
THE NUMBER OF SURVIVING OVA DURING EARLY PREGNANCY IN OLD AND YOUNG HAMSTERS

	corpora lutea per ovary	ova/oviduct at 56 hr	implants/ cornu at 132 hr
Group	Mean ± SE	Mean ± SE	Mean ± SE
Young	7.3 ± 0.14 (12)	6.6 ± 0.1 (12)	6.1 ± 1.0 (10)
Old	6.8 ± 0.19 (10)	4.9 ± 0.2 (10)	2.8 ± 1.5 (12)
Difference	0.5[a]	1.7[b]	3.3[c]

[a] Difference not significant $p > 0.05$ (number of ovaries).

[b] Difference significant $p < 0.01$ (number of oviducts).

[c] Difference significant $p < 0.01$ (number of cornu).

(1966) recovered fewer ovulated ova from the uteri of old mice than from young animals.

The results reported here indicate that the ova are not left behind in the oviduct and some other explanation is needed. Accelerated tubal transport and early entry into the uterus is one possible explanation. The formation of anovulatory corpora lutea in the senescent hamster ovary is another. These structures have been found in the hamster, albeit infrequently (Horowitz, 1967). A detailed histological study of the senescent hamster corpus luteum is needed to clarify this point. Whatever their fate, it is clear that there are fewer ova available for implantation in senescent hamsters.

Failure of implantation has been considered one of the prime reasons for the decrease in fecundity in rodents (Biggers, *et al.*, 1962; Blaha, 1964a, and Thorneycroft and Soderwall, 1969). The results here clearly indicate that the greatest loss of fertilized ova occurs between 56 and 132 hr of development, at the time when implantation normally occurs in hamsters (Ward, 1948).

The reasons for reduced implantation are not clear. Ovariectomized senescent mice maintained on a normal hormonal regimen do not respond to a decidualizing stimulus as efficiently as young animals maintained on the same hormonal scheme (Finn, 1966). Blaha (1967) reported essentially the same results using intact hamsters. Both authors conclude that this is due to a decreased uterine responsiveness and not to deficient hormonal environment.

Blaha (1967) also states that exogenous progesterone does not increase the ability of the senescent uterus to respond to a decidualizing stimulus. The progesterone was given following traumatization of the uterus but not during the normal sensitizing period, therefore, the validity of the conclusion is questionable. PMS, given to senescent hamsters prior to ovulation increased the decidual response (Blaha, 1967). It could be interpreted that the senescent uterus has become refractory to progesterone and requires a higher level of this hormone than is normally needed.

This latter theory is supported by recent work by Blaha (1970). Senescent hamsters given ovarian grafts from young donors were more successful in maintaining transferred blastocysts than intact old animals. This increase is apparently due to an increased number of functional corpora lutea in the grafted ovaries (Blaha, 1970).

The uterine stromal cells proliferate in response to proper hormonal stimulation in ovariectomized senescent mice, but epithelial cells do not respond (Finn and Martin, 1969). In young mice, at the time of implantation, the lateral epithelial surfaces come together so the lumen has a slit-like appearance (Finn and Martin, 1967). Since this closure is associated with increased adhesiveness of the epithelial cells which play an important role in blastocyst attachment (Nilsson, 1966), this would necessarily have a detrimental effect on implantation.

The role of the uterus in reproductive senescence has recently been reviewed by Biggers (1969) and Finn (1970).

Based upon the results of this experiment, the following scheme of embryonic death during early pregnancy which leads to a reduction in litter size is proposed: There may only be slight change in the number of ovulations. There is a 27% decrease in the number of surviving ova prior to implantation, and a 59% decrease in the total number of implants. In addition, there is a decrease in surviving implants by Day 8 of pregnancy and a further reduction, by resorption, of established implants during the latter half of pregnancy (Thorneycroft and Soderwall, 1969).

REFERENCES

BIGGERS, J. D. (1969). Problems concerning the uterine causes of embryonic death, with special reference to the effects of aging of the uterus. *J. Reprod. Fert. Suppl.* **8,** 27–43.

BIGGERS, J. D., FINN, C. A., AND MCLAREN, A. (1962). Long-term reproductive performance of female mice 2. Variation of litter size with parity. *J. Reprod. Fert.* **3,** 313–330.

BLAHA, G. C. (1964a). Reproductive senescence in the female golden hamster. *Anat. Rec.* **150,** 405–412.

BLAHA, G. C. (1964b). Effect of age of the donor and recipient on the development of transferred golden hamster ova. *Anat. Rec.* **150,** 413–416.

BLAHA, G. C. (1967). Effects of age, treatment, and method of induction on deciduomata in the golden hamster. *Fert. Steril.* **18,** 477–485.

BLAHA, G. C. (1970). The influence of ovarian grafts from young donors on the development of transferred ova in aged golden hamsters. *Fert. Steril.* **21,** 268–273.

DIXON, W. J., AND MASSEY, F. J. (1957). "Introduction to Statistical Analysis" 2nd Ed., p. 115. McGraw-Hill Co., New York.

FINN, C. A. (1966). The initiation of the decidual cell reaction in the uterus of the aged mouse. *J. Reprod. Fert.* **11,** 423–428.

FINN, C. A. (1970). The aging uterus and its influence on reproductive capacity. *J. Reprod. Fert. Suppl.* **12,** 31–38.

FINN, C. A., AND MARTIN, L. (1967). Patterns of cell division in the mouse uterus during early pregnancy. *J. Endocrinol.* **39,** 593–597.

FINN, C. A., AND MARTIN, L. (1969). The cellular response of the uterus of the aged mouse to oestrogen and progesterone. *J. Reprod. Fert.* **20,** 545–547.

HARMAN, S. M., AND TALBERT, G. B. (1967). Structural changes in corpora lutea and decline of reproductive function in aging female mice. *Amer. Zool.* **7,** 757–758.

HARVEY, E. B., YANAGIMACHI, R., AND CHANG, M. C. (1961). Onset of estrus and ovulation in the golden hamster. *J. Exp. Zool.* **146,** 231–236.

HOROWITZ, M. (1967). The anovular corpus luteum in the golden hamster (*Mesocricetus auratus* Waterhouse) and comparisons with the normal corpus luteum and follicle of atresia. *Acta Anat.* **66,** 199–225.

INGRAM, D. L., MANDL, A. M., AND ZUCKERMAN, S. (1958). The influence of age on litter size. *J. Endocrinol.* **17,** 280–285.

NILSSON, O. (1966). Estrogen-induced increase of adhesiveness in uterine epithelium of mouse and rat. *Exp. Cell Res.* **43,** 239–241.

ORSINI, M. W. (1961). The external vaginal phenomena characterizing the stages of the estrus cycle, pregnancy, pseudopregnancy, lactation, and the anestrous hamster, *Mesocricetus auratus* Waterhouse. *Proc. Anim. Care Panel* **11,** 193–206.

ORSINI, M. W. (1962a). Technique of preparation, study and photography of benzyl-benzoate cleared material for embryological studies. *J. Reprod. Fert.* **3,** 283–287.

ORSINI, M. W. (1962b). Study of ovo-implantation in the hamster, rat, mouse, guinea pig and rabbit in cleared uterine tracts. *J. Reprod. Fert.* **3,** 288–293.

SODERWALL, A. L. (1945). The normal reproductive behavior of the female hamster. *Proc. Ore. Acad. Sci.* **1,** 45.

SODERWALL, A. L., KENT, H. A., TURBYFILL, C. L., AND BRITENBAKER, A. L. (1960). Variation in gestation length and litter size of the golden hamster, *Mesocricetus auratus. J. Geront.* **15,** 246–248.

TALBERT, G. B., AND KROHN, P. L. (1966). Effect of maternal age on viability of ova and uterine support of pregnancy in mice. *J. Reprod. Fert.* **11,** 399–406.

THORNEYCROFT, I. H., AND SODERWALL, A. L. (1969). The nature of the litter size loss in senescent hamsters. *Anat. Rec.* **165,** 343–348.

WARD, M. C. (1948). The early development and implantation of the golden hamster, *Cricetus auratus*, and the associated endometrial changes. *Amer. J. Anat.* **82,** 231–262.